Employee Performance Mapping

Output screenshots (MySQL 8.0)

Q1 (a): Creating the employee database and checking the imported tables

```
1  CREATE DATABASE employee;
2  USE employee;
3
4  SHOW TABLES;
```

Result Grid

Tables_in_employee
data_science_team
emp_record_table
proj_table

Q1 (b): emp_record_table after import

```
1  SELECT * FROM emp_record_table;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
E001	Arthur	Black	M	PRESIDENT	ALL	20	USA	NORTH AMERICA	16500	5	NULL	NULL
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE	9000	2	E428	P204
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA	5500	5	E083	P103
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA	7700	1	E083	P302
E083	Patrick	Voltz	M	MANAGER	HEALTHCARE	15	USA	NORTH AMERICA	9500	5	E001	NULL
E103	Emily	Grove	F	MANAGER	FINANCE	14	CANADA	NORTH AMERICA	10500	4	E001	NULL
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE	7500	5	E428	P204
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA	6500	2	E583	P109
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA	7000	3	E583	NA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA	5000	3	E103	P105
E428	Pete	Allen	M	MANAGER	AUTOMOTIVE	14	GERMANY	EUROPE	11000	4	E001	NULL
E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA	4000	4	E583	P109
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA	5000	2	E083	P103
E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE	4300	1	E428	P204
E583	Janet	Hale	F	MANAGER	RETAIL	14	COLOMBIA	SOUTH AMERICA	10000	2	E001	NULL
E612	Tracy	Norris	F	MANAGER	RETAIL	13	INDIA	ASIA	8500	4	E001	NULL
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA	3000	1	E612	P406
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA	2800	4	E612	P406

Q1 (c): proj_table after import

```
1  SELECT * FROM proj_table;
```

Result Grid

PROJECT_ID	PROJ_NAME	DOMAIN	START _DATE	CLOSURE_DATE	DEV_QTR	STATUS
P103	Drug Discovery	HEALTHCARE	04-06-2021	6/20/2021	Q1	DONE
P105	Fraud Detection	FINANCE	04-11-2021	6/25/2021	Q1	DONE
P109	Market Basket Analysis	RETAIL	04-12-2021	6/30/2021	Q1	DELAYED
P204	Supply Chain Management	AUTOMOTIVE	07/15/2021	9/28/2021	Q2	WIP
P302	Early Detection of Lung Cancer	HEALTHCARE	10-08-2021	12/18/2021	Q3	YTS
P406	Customer Sentiment Analysis	RETAIL	07-09-2021	9/24/2021	Q2	WIP

Q1 (d): data_science_team after import

```
1  SELECT * FROM data_science_team;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT
E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA
E010	William	Butler	M	LEAD DATA SCIENTIST	AUTOMOTIVE	12	FRANCE	EUROPE
E052	Dianna	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	6	CANADA	NORTH AMERICA
E057	Dorothy	Wilson	F	SENIOR DATA SCIENTIST	HEALTHCARE	9	USA	NORTH AMERICA
E204	Karene	Nowak	F	SENIOR DATA SCIENTIST	AUTOMOTIVE	8	GERMANY	EUROPE
E245	Nian	Zhen	M	SENIOR DATA SCIENTIST	RETAIL	6	CHINA	ASIA
E260	Roy	Collins	M	SENIOR DATA SCIENTIST	RETAIL	7	INDIA	ASIA
E403	Steve	Hoffman	M	ASSOCIATE DATA SCIENTIST	FINANCE	4	USA	NORTH AMERICA
E478	David	Smith	M	ASSOCIATE DATA SCIENTIST	RETAIL	3	COLOMBIA	SOUTH AMERICA
E505	Chad	Wilson	M	ASSOCIATE DATA SCIENTIST	HEALTHCARE	5	CANADA	NORTH AMERICA
E532	Claire	Brennan	F	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	3	GERMANY	EUROPE
E620	Katrina	Allen	F	JUNIOR DATA SCIENTIST	RETAIL	2	INDIA	ASIA
E640	Jenifer	Jhones	F	JUNIOR DATA SCIENTIST	RETAIL	1	COLOMBIA	SOUTH AMERICA

Q2: Employee list with department details

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT
FROM emp_record_table
ORDER BY DEPT, EMP_ID;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT
E001	Arthur	Black	M	ALL
E010	William	Butler	M	AUTOMOTIVE
E204	Karene	Nowak	F	AUTOMOTIVE
E428	Pete	Allen	M	AUTOMOTIVE
E532	Claire	Brennan	F	AUTOMOTIVE
E005	Eric	Hoffman	M	FINANCE
E103	Emily	Grove	F	FINANCE
E403	Steve	Hoffman	M	FINANCE
E052	Dianna	Wilson	F	HEALTHCARE
E057	Dorothy	Wilson	F	HEALTHCARE
E083	Patrick	Voltz	M	HEALTHCARE
E505	Chad	Wilson	M	HEALTHCARE
E245	Nian	Zhen	M	RETAIL
E260	Roy	Collins	M	RETAIL
E478	David	Smith	M	RETAIL
E583	Janet	Hale	F	RETAIL
E612	Tracy	Norris	F	RETAIL
E620	Katrina	Allen	F	RETAIL
E640	Jenifer	Jhones	F	RETAIL

Q3 (a): Employees with EMP_RATING less than 2

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING < 2;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
E057	Dorothy	Wilson	F	HEALTHCARE	1
E532	Claire	Brennan	F	AUTOMOTIVE	1
E620	Katrina	Allen	F	RETAIL	1

Q3 (b): Employees with EMP_RATING between 2 and 4

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING BETWEEN 2 AND 4;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
E005	Eric	Hoffman	M	FINANCE	3
E010	William	Butler	M	AUTOMOTIVE	2
E103	Emily	Grove	F	FINANCE	4
E245	Nian	Zhen	M	RETAIL	2
E260	Roy	Collins	M	RETAIL	3
E403	Steve	Hoffman	M	FINANCE	3
E428	Pete	Allen	M	AUTOMOTIVE	4
E478	David	Smith	M	RETAIL	4
E505	Chad	Wilson	M	HEALTHCARE	2
E583	Janet	Hale	F	RETAIL	2
E612	Tracy	Norris	F	RETAIL	4
E640	Jenifer	Jhones	F	RETAIL	4

Q3 (c): Employees with EMP_RATING greater than 4

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, GENDER, DEPT AS DEPARTMENT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING > 4;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	GENDER	DEPARTMENT	EMP_RATING
E001	Arthur	Black	M	ALL	5
E052	Dianna	Wilson	F	HEALTHCARE	5
E083	Patrick	Voltz	M	HEALTHCARE	5
E204	Karene	Nowak	F	AUTOMOTIVE	5

Q4: Employees in leadership positions (Manager, President, CEO)

```sql
SELECT EMP_ID, FIRST_NAME, ROLE, DEPT AS DEPARTMENT
FROM emp_record_table
WHERE ROLE IN ('MANAGER', 'PRESIDENT', 'CEO')
ORDER BY ROLE;
```

Result Grid

EMP_ID	FIRST_NAME	ROLE	DEPARTMENT
E083	Patrick	MANAGER	HEALTHCARE
E103	Emily	MANAGER	FINANCE
E428	Pete	MANAGER	AUTOMOTIVE
E583	Janet	MANAGER	RETAIL
E612	Tracy	MANAGER	RETAIL
E001	Arthur	PRESIDENT	ALL

Q5: Employee ratings with the max rating of each department

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT AS DEPARTMENT, EMP_RATING,
       MAX(EMP_RATING) OVER (PARTITION BY DEPT) AS MAX_DEPT_RATING
FROM emp_record_table
ORDER BY DEPT, EMP_RATING DESC;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPARTMENT	EMP_RATING	MAX_DEPT_RATING
E001	Arthur	Black	PRESIDENT	ALL	5	5
E204	Karene	Nowak	SENIOR DATA SCIENTIST	AUTOMOTIVE	5	5
E428	Pete	Allen	MANAGER	AUTOMOTIVE	4	5
E010	William	Butler	LEAD DATA SCIENTIST	AUTOMOTIVE	2	5
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1	5
E103	Emily	Grove	MANAGER	FINANCE	4	4
E005	Eric	Hoffman	LEAD DATA SCIENTIST	FINANCE	3	4
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	FINANCE	3	4
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	5	5
E083	Patrick	Voltz	MANAGER	HEALTHCARE	5	5
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	HEALTHCARE	2	5
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	1	5
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	RETAIL	4	4
E612	Tracy	Norris	MANAGER	RETAIL	4	4
E478	David	Smith	ASSOCIATE DATA SCIENTIST	RETAIL	4	4
E260	Roy	Collins	SENIOR DATA SCIENTIST	RETAIL	3	4
E583	Janet	Hale	MANAGER	RETAIL	2	4
E245	Nian	Zhen	SENIOR DATA SCIENTIST	RETAIL	2	4
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	RETAIL	1	4

Q6: Minimum and maximum salary by role

```sql
SELECT ROLE,
       MIN(SALARY) AS MIN_SALARY,
       MAX(SALARY) AS MAX_SALARY
FROM emp_record_table
GROUP BY ROLE
ORDER BY MAX_SALARY DESC;
```

Result Grid

ROLE	MIN_SALARY	MAX_SALARY
PRESIDENT	16500	16500
MANAGER	8500	11000
LEAD DATA SCIENTIST	8500	9000
SENIOR DATA SCIENTIST	5500	7700
ASSOCIATE DATA SCIENTIST	4000	5000
JUNIOR DATA SCIENTIST	2800	3000

Q7: Ranking employees by years of experience

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, EXP,
       RANK() OVER (ORDER BY EXP DESC) AS EXP_RANK,
       DENSE_RANK() OVER (ORDER BY EXP DESC) AS EXP_DENSE_RANK
FROM emp_record_table;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	EXP	EXP_RANK	EXP_DENSE_RANK
E001	Arthur	Black	PRESIDENT	20	1	1
E083	Patrick	Voltz	MANAGER	15	2	2
E103	Emily	Grove	MANAGER	14	3	3
E583	Janet	Hale	MANAGER	14	3	3
E428	Pete	Allen	MANAGER	14	3	3
E612	Tracy	Norris	MANAGER	13	6	4
E010	William	Butler	LEAD DATA SCIENTIST	12	7	5
E005	Eric	Hoffman	LEAD DATA SCIENTIST	11	8	6
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	9	9	7
E204	Karene	Nowak	SENIOR DATA SCIENTIST	8	10	8
E260	Roy	Collins	SENIOR DATA SCIENTIST	7	11	9
E245	Nian	Zhen	SENIOR DATA SCIENTIST	6	12	10
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	6	12	10
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	5	14	11
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	4	15	12
E478	David	Smith	ASSOCIATE DATA SCIENTIST	3	16	13
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	3	16	13
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	2	18	14
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	1	19	15

Q8: View of employees with salary more than 6000

```sql
CREATE VIEW emp_salary_above_6000 AS
SELECT EMP_ID, FIRST_NAME, LAST_NAME, COUNTRY, SALARY
FROM emp_record_table
WHERE SALARY > 6000;

SELECT * FROM emp_salary_above_6000
ORDER BY COUNTRY, SALARY DESC;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	COUNTRY	SALARY
E103	Emily	Grove	CANADA	10500
E245	Nian	Zhen	CHINA	6500
E583	Janet	Hale	COLOMBIA	10000
E010	William	Butler	FRANCE	9000
E428	Pete	Allen	GERMANY	11000
E204	Karene	Nowak	GERMANY	7500
E612	Tracy	Norris	INDIA	8500
E260	Roy	Collins	INDIA	7000
E001	Arthur	Black	USA	16500
E083	Patrick	Voltz	USA	9500
E005	Eric	Hoffman	USA	8500
E057	Dorothy	Wilson	USA	7700

Q9: Country wise view of employees earning more than 6000

```sql
CREATE VIEW country_wise_high_earners AS
SELECT COUNTRY,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       GROUP_CONCAT(CONCAT(FIRST_NAME, ' ', LAST_NAME)
                    ORDER BY SALARY DESC SEPARATOR ', ') AS EMPLOYEES
FROM emp_record_table
WHERE SALARY > 6000
GROUP BY COUNTRY;

SELECT * FROM country_wise_high_earners
ORDER BY NO_OF_EMPLOYEES DESC;
```

Result Grid

COUNTRY	NO_OF_EMPLOYEES	EMPLOYEES
USA	4	Arthur Black, Patrick Voltz, Eric Hoffman, Dorothy Wilson
GERMANY	2	Pete Allen, Karene Nowak
INDIA	2	Tracy Norris, Roy Collins
CANADA	1	Emily Grove
CHINA	1	Nian Zhen
COLOMBIA	1	Janet Hale
FRANCE	1	William Butler

Q10 (a): Execution plan before creating the index

```
1  EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Result Grid

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	emp_record_table	NULL	ALL	NULL	NULL	NULL	NULL	19	10.0	Using where

Q10 (b): Creating the index and the execution plan after

```
1  CREATE INDEX idx_first_name ON emp_record_table (FIRST_NAME(20));
2
3  EXPLAIN SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Result Grid

id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	emp_record_table	NULL	ref	idx_first_name	idx_first_name	83	const	1	100.0	Using where

Q10 (c): Result of the lookup for Eric

```
1  SELECT * FROM emp_record_table WHERE FIRST_NAME = 'Eric';
```

Result Grid

	EMP_ID	FIRST_NAME	LAST_NAME	GENDER	ROLE	DEPT	EXP	COUNTRY	CONTINENT	SALARY	EMP_RATING	MANAGER_ID	PROJ_ID
▶	E005	Eric	Hoffman	M	LEAD DATA SCIENTIST	FINANCE	11	USA	NORTH AMERICA	8500	3	E103	P105

Q11 (a): Average salary by continent and country

```sql
SELECT CONTINENT, COUNTRY,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       ROUND(AVG(SALARY), 2) AS AVG_SALARY
FROM emp_record_table
GROUP BY CONTINENT, COUNTRY
ORDER BY CONTINENT, AVG_SALARY DESC;
```

Result Grid

CONTINENT	COUNTRY	NO_OF_EMPLOYEES	AVG_SALARY
ASIA	CHINA	1	6500.00
ASIA	INDIA	3	6166.67
EUROPE	FRANCE	1	9000.00
EUROPE	GERMANY	3	7600.00
NORTH AMERICA	USA	5	9440.00
NORTH AMERICA	CANADA	3	7000.00
SOUTH AMERICA	COLOMBIA	3	5600.00

Q11 (b): Average salary by continent

```sql
SELECT CONTINENT,
       COUNT(EMP_ID) AS NO_OF_EMPLOYEES,
       ROUND(AVG(SALARY), 2) AS AVG_SALARY
FROM emp_record_table
GROUP BY CONTINENT
ORDER BY AVG_SALARY DESC;
```

Result Grid

CONTINENT	NO_OF_EMPLOYEES	AVG_SALARY
NORTH AMERICA	8	8525.00
EUROPE	4	7950.00
ASIA	4	6250.00
SOUTH AMERICA	3	5600.00

Q12 (a): Bonus per employee (5% of salary x rating)

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, SALARY, EMP_RATING,
       ROUND(0.05 * SALARY * EMP_RATING, 2) AS BONUS
FROM emp_record_table
ORDER BY BONUS DESC;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	SALARY	EMP_RATING	BONUS
E001	Arthur	Black	PRESIDENT	16500	5	4125.00
E083	Patrick	Voltz	MANAGER	9500	5	2375.00
E428	Pete	Allen	MANAGER	11000	4	2200.00
E103	Emily	Grove	MANAGER	10500	4	2100.00
E204	Karene	Nowak	SENIOR DATA SCIENTIST	7500	5	1875.00
E612	Tracy	Norris	MANAGER	8500	4	1700.00
E052	Dianna	Wilson	SENIOR DATA SCIENTIST	5500	5	1375.00
E005	Eric	Hoffman	LEAD DATA SCIENTIST	8500	3	1275.00
E260	Roy	Collins	SENIOR DATA SCIENTIST	7000	3	1050.00
E583	Janet	Hale	MANAGER	10000	2	1000.00
E010	William	Butler	LEAD DATA SCIENTIST	9000	2	900.00
E478	David	Smith	ASSOCIATE DATA SCIENTIST	4000	4	800.00
E403	Steve	Hoffman	ASSOCIATE DATA SCIENTIST	5000	3	750.00
E245	Nian	Zhen	SENIOR DATA SCIENTIST	6500	2	650.00
E640	Jenifer	Jhones	JUNIOR DATA SCIENTIST	2800	4	560.00
E505	Chad	Wilson	ASSOCIATE DATA SCIENTIST	5000	2	500.00
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	7700	1	385.00
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	4300	1	215.00
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	3000	1	150.00

Q12 (b): Total bonus cost

```sql
SELECT ROUND(SUM(0.05 * SALARY * EMP_RATING), 2) AS TOTAL_BONUS_COST
FROM emp_record_table;
```

Result Grid

TOTAL_BONUS_COST
23985.00

Q12 (c): Employees flagged for training (rating below 2)

```sql
SELECT EMP_ID, FIRST_NAME, LAST_NAME, ROLE, DEPT, EMP_RATING
FROM emp_record_table
WHERE EMP_RATING < 2;
```

Result Grid

EMP_ID	FIRST_NAME	LAST_NAME	ROLE	DEPT	EMP_RATING
E057	Dorothy	Wilson	SENIOR DATA SCIENTIST	HEALTHCARE	1
E532	Claire	Brennan	ASSOCIATE DATA SCIENTIST	AUTOMOTIVE	1
E620	Katrina	Allen	JUNIOR DATA SCIENTIST	RETAIL	1